

DR. MOHD SAKIB

Assistant Professor, School of Computer Science
UPES University (University of Petroleum and Energy Studies), Dehradun, India

Research Focus: Deep Learning, Ensemble Learning, Predictive Analytics, Time-Series Forecasting, and Explainable AI, with a focus on Data-Driven Modelling and Decision-Making across diverse application domains.

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Google Scholar: scholar.google.com/citations?user=MvU637IAAAAJ **LinkedIn:** linkedin.com/in/sakib9286

Google Scholar 235+ Citations · h-index 7	5 SCIE-Indexed Journal Articles	IEEE Member UGC NET Qualified
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Bibliometric indicators as of September 2026.

PROFESSIONAL SUMMARY

Computer scientist working at the intersection of deep learning and time-series analysis, holding a Ph.D. from Aligarh Muslim University (2024) and more than five years of research experience. Research contributions centre on decomposition-driven and ensemble deep learning architectures for nonlinear, non-stationary temporal dynamics across energy demand, smart grids, and epidemiological surveillance, published in five SCIE-indexed journals (four Q1) alongside IEEE conference proceedings and book chapters with CRC Press and IGI Global. Co-editor of two published IGI Global volumes, with further volumes under proposal (CRC Press, Elsevier) and a graduate textbook in preparation. Serves as Principal Investigator on an externally funded assistive-technology project that has produced granted registered designs in the United Kingdom and India. Reviews for eight international journals and supervises undergraduate and postgraduate research through to peer-reviewed publication.

EDUCATION

Ph.D. in Computer Science Aligarh Muslim University, Aligarh, India Supervisor: Prof. Suhel Mustajab Thesis: An Efficient Ensemble Deep Learning Model for Estimation and Forecasting of Time Series Data	April 2021 – December 2024
Master of Computer Applications Aligarh Muslim University, Aligarh, India Dissertation: Mess Management System: A Case Study of SS Hall North CGPA: 7.93/10	July 2015 – December 2018
Bachelor of Computer Applications Aligarh Muslim University, Aligarh, India Dissertation: Design and Development of an Online Complaint Management System Percentage: 75.6%	June 2012 – July 2015

CURRENT POSITION

Assistant Professor School of Computer Science UPES University (University of Petroleum and Energy Studies), Dehradun, India – 248007	June 2025 – Present
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EXPERIENCE

Assistant Professor Department of Computer Applications Madanapalle Institute of Technology and Science (MITS Deemed to be University) Madanapalle, Andhra Pradesh, India – 517325	August 2024 – June 2025
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Centre of Advanced Research in Electrified Transportation (CARET)

AMU Research Centre, Aligarh, Uttar Pradesh, India – 202002

Project: Machine learning-based day-ahead market energy usage bidding for smart microgrids

ACHIEVEMENTS & AWARDS

- Qualified the UGC NET Examination in Computer Science twice (December 2018 and December 2019)
- Member, IEEE and IEEE Computer Society; Life Member, Indian Society for Technical Education (ISTE), since 2025
- UGC Non-NET Fellowship (University Grants Commission, India) for doctoral research, held throughout the Ph.D. programme; University Merit Scholarship (2017–2018)
- Two appreciation letters from the Head of Department, MITS, for excellence in teaching (student feedback 4.70/5 and 4.77/5, 2024–25)
- Springer Nature reviewer certificates for eleven manuscripts in 2026: *Scientific Reports* (3), *Cluster Computing* (4), *Signal, Image and Video Processing* (2), *Journal of Big Data* (1), and *Discover Artificial Intelligence* (1)
- Certificates of appreciation as Reviewer and as Presenter, 2026 International Conference on Advancement in Cyber Security and Digital Forensics (ICACSDF), UPES, 11–12 September 2026
- Jury Member, Smart India Hackathon 2026 (Internal Round), UPES Dehradun, August–September 2026; a Ministry of Education, Government of India, initiative

RESEARCH GRANTS & PROJECT FUNDING

E-Vision: An Intelligent Assistive Application for the Visually Impaired

September 2025

Role: Principal Investigator (PI)

Funding Agency: Cadre Tech Innovative Solutions LLC, Delaware, USA

Reference ID: CT_US250102

Sanctioned Amount: INR 2,00,000 **Status:** Approved for funding

Host Institution: School of Computer Science (SOCS), UPES University, Dehradun, India

Outcome: UK Registered Design No. 6538179, *AI Enabled Wearable Device for Blind Individuals*, granted 16 July 2026; and Indian Registered Design No. 510366-001, *AI-Enabled Wearable Device for Visually Impaired Persons*, registered 22 July 2026

RESEARCH PUBLICATIONS

Journal Articles (Published)

- **Sakib, M., & Mustajab, S.** *CEEMDAN-TCN-AutoLSTM: A Triple-Force Ensemble Using Cross-Stitch Networks for Short-Term Load Forecasting*. *Arabian Journal for Science and Engineering* (2025). [SCIE, Scopus; IF: 2.9; Q1]. [doi:10.1007/s13369-025-10398-1](https://doi.org/10.1007/s13369-025-10398-1)
- **Sakib, M., Mustajab, S., & Alam, M.** *Ensemble deep learning techniques for time series analysis: a comprehensive review, applications, open issues, challenges, and future directions*. *Cluster Computing*, 28(1) (2025). [SCIE, Scopus; IF: 3.6; Q1]. [doi:10.1007/s10586-024-04684-0](https://doi.org/10.1007/s10586-024-04684-0)
- **Sakib, M., Siddiqui, T., Mustajab, S., Alotaibi, R. M., Alshareef, N. M., & Khan, M. Z.** *An ensemble deep learning framework for energy demand forecasting using genetic algorithm-based feature selection*. *PLOS One*, 20(1) (2025). [SCIE, Scopus; IF: 3.7; Q1]. [doi:10.1371/journal.pone.0310465](https://doi.org/10.1371/journal.pone.0310465)
- **Sakib, M., Ahmad, S., Anwar, K., & Saqib, M.** *Optimizing support vector regression using grey wolf optimizer for enhancing energy efficiency and building prototype architecture*. *Cluster Computing*, 28(1), 60 (2025). [SCIE, Scopus; IF: 3.6; Q1]. [doi:10.1007/s10586-024-04802-y](https://doi.org/10.1007/s10586-024-04802-y)
- **Sakib, M., & Siddiqui, T.** *Multi-network-based ensemble deep learning model to forecast Ross River virus outbreak in Australia*. *International Journal of Pattern Recognition and Artificial Intelligence*, 37(10) (2023). [SCIE, Scopus; IF: 1.5; Q3]. [doi:10.1142/S0218001423520158](https://doi.org/10.1142/S0218001423520158)

- **Sakib, M., & Mustajab, S.** *Enhanced multi-variate time series prediction through statistical-deep learning integration: the VAR-stacked LSTM model*. SN Computer Science, 5(5), 573 (2024). [Scopus]. doi:10.1007/s42979-024-02950-x

Conference Proceedings (Published)

- **Sakib, M., Mustajab, S., & Siddiqui, T.** *Deep learning-based heartbeat classification of 12-lead ECG time series signal*. In *2023 4th International Conference on Data Analytics for Business and Industry (ICDABI-2023)*, pp. 273–278. IEEE. [Scopus]. doi:10.1109/ICDABI60145.2023.10629531
- **Sakib, M., & Siddiqui, T.** *Anomaly detection of ECG time series signal using autoencoders neural network*. In *2023 7th International Conference on Computing, Communication, Control and Automation (ICCUBEA)*, pp. 1–7. IEEE. [Scopus]. doi:10.1109/ICCUBEA58933.2023.10392094
- **Sakib, M., Mustajab, S., Anwar, K., & Alam, M.** *BiLSTM-based approach for energy demand forecasting and synchronization among smart grids*. In *2024 IEEE Third International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES)*, pp. 969–974. IEEE. [Scopus]. doi:10.1109/ICPEICES62430.2024.10719085
- **Khan, M., Sakib, M., Siddiqui, T., & Awasthi, S.** *Long-horizon forecasting for sensor systems: an experimental study using linear and transformer-based models*. In *2026 Second International Conference on Emerging Computational Intelligence (ICECI)*, pp. 1–6. IEEE. [Scopus]. doi:10.1109/ICECI69159.2026.11519399

Book Chapters (Published)

- **Saqib, M., Sahu, S. A., Sakib, M., & Al-Ammar, E. A.** *Machine learning-based day-ahead market energy usage bidding for smart microgrids*. In *Electric Vehicle Integration in a Smart Microgrid Environment*, pp. 249–266. CRC Press (Taylor & Francis) (2021). [Scopus]. doi:10.1201/9780367423926-10
- **Suganya, S., Sakib, M., & Malini, M.** *Mitigating adversarial attacks in industrial cybersecurity through NLP strategies for e-commerce: challenges and emerging trends*. In *AI-Driven Security and Intelligence in Cloud and Internet of Things Systems*. IGI Global (2025). doi:10.4018/979-8-3373-7503-8.ch003

Edited Books (Published)

- *AI Advancements in Internet of Things, Smart Healthcare, and Intelligent Devices*. Edited by Muneer Khan, Shantanu Awasthi, **Mohd Sakib**, and Mohd Vaseem Khan. IGI Global (2025). ISBN13: 9798337357270 | EISBN13: 9798337357294. doi:10.4018/979-8-3373-5727-0
- *AI-Driven Security and Intelligence in Cloud and Internet of Things Systems*. Edited by **Mohd Sakib**, Muneer Khan, Shantanu Awasthi, and Mohd Vaseem Khan. IGI Global (2025). ISBN13: 9798337375038 | EISBN13: 9798337375052. doi:10.4018/979-8-3373-7503-8

Authored Textbook (In Preparation)

- **Sakib, M.** *Foundation Models and the Structure of Time: An Independent Benchmark of Pretrained, Decomposition, and Ensemble Forecasting*. Single-authored graduate textbook, fifteen chapters (approx. 375 pages), with companion benchmark code and course materials. **Status:** In preparation; proposal to be submitted to Oxford University Press.

Edited Volumes in Progress

- *Artificial Intelligence for Industry 5.0 Engineering Systems: Deep Learning, IoT, and Intelligent Automation*. Edited by Prof. Vinay Rishiwal, Muneer Khan, **Mohd Sakib**, and Prof. Ashutosh Kumar Bhatt. CRC Press (Taylor & Francis Group). **Status:** Proposal under review; revised proposal submitted at the editors' request.
- *Intelligent IoT: AI-Driven Sensing, Connectivity, and Control*. Edited by Muneer Khan and **Mohd Sakib**. Elsevier. Eighteen commissioned chapters planned. **Status:** Proposal submitted.

Accepted / Forthcoming

- **Sakib, M., Khan, M., & Awasthi, S.** *Explainable self-selecting ensemble deep learning for sustainable multi-domain forecasting*. *5th Elsevier International Conference on Machine Learning & Data Engineering (ICMLDE 5.0)*, Dehradun, India, 27–28 November 2026. **Accepted;** to appear in *Procedia Computer Science* (Elsevier). [Scopus]

- Ranjan, K., Garg, K., & **Sakib, M.** *Security-aware multi-view graph contrastive learning for network intrusion detection*. 2026 International Conference on Advancement in Cyber Security and Digital Forensics (ICACSDF), UPES, Dehradun, 11–12 September 2026. **Presented.**
- **Sakib, M.**, & Khan, M. *A hybrid soft computing framework for intelligent time-series forecasting using STL decomposition and Bayesian-optimized deep learning*. Book chapter in *Soft Computing for Web Intelligence: Theory and Applications*, CRC Press (Taylor & Francis Group). **Accepted.**

Manuscripts Under Review

- Singh, K., Agarawal, P., & **Sakib, M.** *Artificial intelligence techniques for cybersecurity: a comprehensive systematic review of methods, applications, challenges, and research directions*. *Journal of Information Security and Applications* (Elsevier). **Under review.**
- **Sakib, M.**, Fazil, M., Qamar, S., & Khan, M. *E³-TSF: A dynamic gated multi-expert transformer framework for robust multi-domain time series forecasting*. *Scientific Reports* (Nature Portfolio). **Under review.**

INTELLECTUAL PROPERTY

Registered Designs

- **UK Registered Design No. 6538179:** *AI Enabled Wearable Device for Blind Individuals*. Registration date: 09 July 2026; grant date: 16 July 2026. International Design Classification: Class 16 (Photographic, Cinematographic and Optical Apparatus), Subclass 06 (Optical Articles). Proprietors: Muneer Khan, **Mohd Sakib**, and Vinay Rishiwal. **Status:** Granted.
- **Indian Registered Design No. 510366-001:** *AI-Enabled Wearable Device for Visually Impaired Persons*. Registration date: 22 July 2026; certificate of registration issued 03 September 2026 by the Patent Office, Government of India (Designs Act, 2000). International Design Classification: Class 16 (Photographic, Cinematographic and Optical Apparatus), Subclass 06 (Optical Articles). Proprietors: Muneer Khan, **Mohd Sakib**, and Ashutosh Kumar Bhatt. **Status:** Registered.
- **UK Design Application No. 6545206:** *A Low-Power On-Device Keyword-Spotting System for AI-Powered Patient Assistance Using TinyML*. Filing date: 07 August 2026. Applicants: Muneer Khan, **Mohd Sakib**, Mridul Pandey, Rajeev Kumar, and Vinay Rishiwal. **Status:** Filed; under examination.

Software Copyright

- *ChronoAnom-X: Explainable Temporal Intelligence Software for Time-Series Anomaly Discovery*. Computer software (Python; pandas, NumPy, scikit-learn, statsmodels, Matplotlib). Authors: **Mohd Sakib** and Muneer Khan. Filed through the Intellectual Property Rights cell, UPES University, Dehradun. **Status:** Under filing.

BRIEF RESEARCH SUMMARY

Doctoral Research

An Efficient Ensemble Deep Learning Model for Estimation and Forecasting of Time Series Data

Led this programme of work end to end, from formulation of the research problem through model design and implementation, experimental evaluation, and preparation of the resulting manuscripts.

- Proposed a decomposition-driven hybrid forecasting framework integrating CEEMDAN with deep temporal convolution and sequence learning architectures to model complex, nonlinear, and non-stationary time series dynamics.
- Developed a VAR–LSTM fusion model combining statistical vector autoregression with deep sequence modelling to capture both linear and nonlinear interdependencies in multivariate time series data.
- Designed a genetic algorithm-optimized ensemble model for dynamic feature selection and forecasting performance enhancement under volatile temporal conditions.
- Investigated attention mechanisms, autoencoder-based sequence modelling, and hierarchical deep learning structures for capturing long-term temporal dependencies and latent representations in sequential data.

Current Research

E³-TSF: A Dynamic Gating Ensemble Transformer Framework for Multi-Domain Time Series Forecasting

- Designing a context-aware ensemble transformer architecture that unifies temporal patch learning and variable-wise interdependence modelling within a dynamic gating framework. The model employs a learnable gating network to adaptively allocate weights between complementary transformer backbones according to local time-series characteristics such as trend behaviour, volatility, and cross-variable interaction.
- Evaluating the framework across heterogeneous domains: energy (ETTh1), traffic (METR-LA), finance (Exchange Rate), and epidemiological forecasting (ILI), demonstrating consistent improvements over standalone transformer models and static ensemble approaches.

STUDENT SUPERVISION & MENTORING

- Supervise undergraduate and postgraduate research at UPES University (previously MITS), from problem formulation and experimental design to manuscript preparation and peer-reviewed publication; four student researchers have co-authored papers accepted or under review at international venues.
- Guided Kushagra Ranjan and Kavya Garg (UPES) on multi-view graph contrastive learning for network intrusion detection; resulting paper presented at the *2026 International Conference on Advancement in Cyber Security and Digital Forensics (ICACSDF)*, September 2026.
- Guided Krishnanand Singh and Pratik Agarawal on a systematic review of artificial intelligence techniques for cybersecurity; manuscript under review at *Journal of Information Security and Applications* (Elsevier).
- Mentor student teams in reproducible experimental practice, scientific writing, and responding to peer review.

TEACHING EXPERIENCE

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| • Data Structures and Algorithms, Theory/Lab – MCA 1st Sem 2024 | • Design and Analysis of Algorithms – B.Sc. 3rd Sem 2025 |
| • Database Management Systems Laboratory – MCA 1st Sem 2024 | • Linux Laboratory – B.Tech. 1st Sem 2025 |
| • Deep Learning – MCA 3rd Sem 2024 | • Data Structures and Algorithms – B.Tech. 2nd Sem 2026 |
| • Artificial Intelligence, Theory/Lab – MCA 2nd Sem 2025 | • Statistics and Analysis – B.Tech. 6th Sem 2026 |

ACADEMIC ACTIVITIES

- Worked as **Criterion In-Charge** for National Board of Accreditation (NBA) documentation and compliance activities.
- Appointed **Students' Holistic Activities Coordinator** (September 2024) and **AICTE-PARAKH Coordinator** (December 2024), Department of Computer Applications, MITS.
- **Co-Convenor**, five-day online Faculty Development Programme (FDP) on “Empowering Minds through AI: Using Innovative Strategies for Effective Problem Solving”.
- **Organizer**, five-day national-level online Faculty Development Programme (FDP) on “Innovative Applications of Artificial Intelligence in Cybersecurity”.
- **Member of Editorial Board & Technical Reviewer**, book titled *Advancing Cybersecurity in Smart Factories Through Autonomous Robotic Defenses*, IGI Global Scientific Publishing, February 2025. ISBN: 979-8-3373-0583-7.
- Attended short-term courses, workshops, and FDPs on time-series forecasting and deep learning.

TECHNICAL SKILLS

SKILL TYPE	SKILL NAME
Programming Languages	Python and SQL
Deep Learning Frameworks	PyTorch, TensorFlow, Keras, and Hugging Face Transformers
Machine Learning & Scientific Computing	scikit-learn, NumPy, pandas, SciPy, and statsmodels
Time-Series Modelling	CEEMDAN/EEMD decomposition, STL, VAR, ARIMA, TCN, LSTM/BiLSTM, and Transformer architectures
Optimization & Hyperparameter Tuning	Genetic Algorithms, Grey Wolf Optimizer, and Bayesian optimization
Visualization	Matplotlib
Databases & Environment	MySQL, Git, $\text{\LaTeX}$, and Linux

REVIEWING SERVICES

- **Journals:** Computers in Biology and Medicine (Elsevier); Scientific Reports (Nature Portfolio); Cluster Computing, Signal, Image and Video Processing, Electrical Engineering, Journal of Big Data, and Discover Artificial Intelligence (Springer); International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems (World Scientific).
- **Conferences:** Technical Program Committee Member, ICIESTR-2024, IIRA 5.0, and ICECI 2026; Reviewer, CICTN-2025, ICDT-2025, and ICACSDF 2026.

REFERENCES

Prof. Tamanna Siddiqui Professor, Department of Computer Science Aligarh Muslim University, Aligarh, India t.siddiqui.cs@amu.ac.in	Prof. Suhel Mustajab Professor, Department of Computer Science Aligarh Muslim University, Aligarh, India s.mustajab.cs@amu.ac.in suhelmustajab@gmail.com	Dr. Mohd Fazil Assistant Professor, College of Computer and Information Sciences Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh, Saudi Arabia mfazil@imamu.edu.sa
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